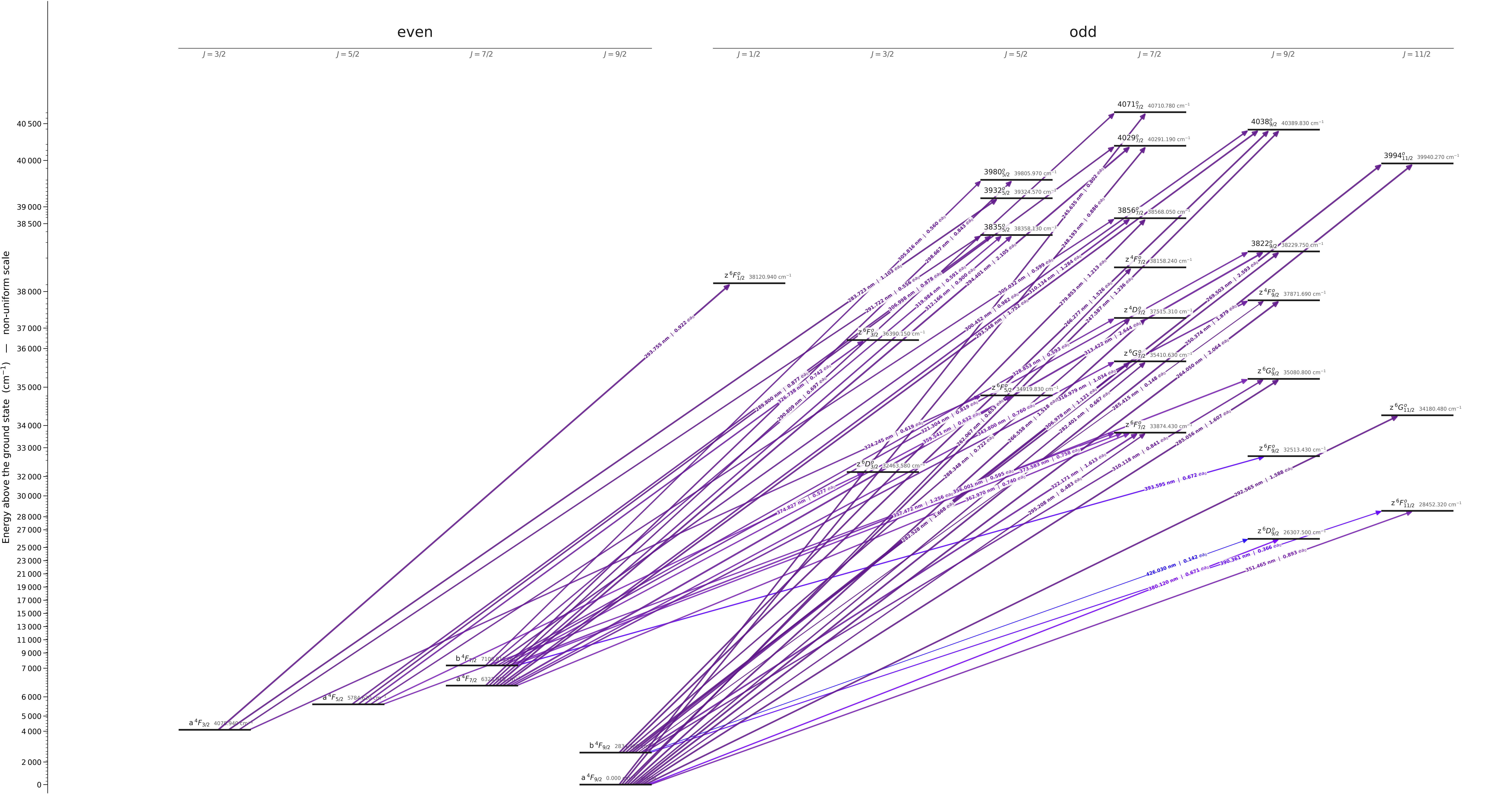
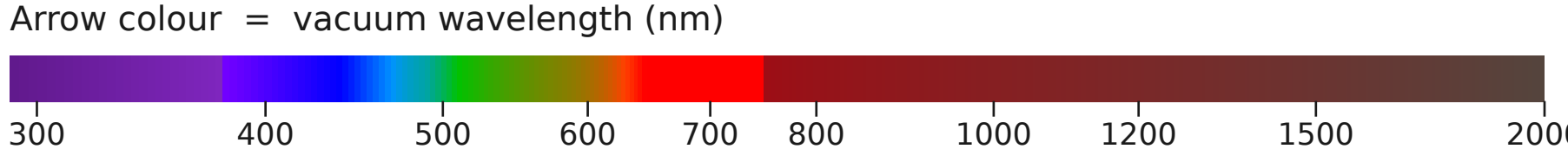


Iridium

Ir (Ir I). Strongest classified transitions below 2 μm (NIST): 29 levels, 53 lines.

Ir⁺ ionisation limit 72323.90 cm⁻¹ = 8.96702 eV (138.267 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle U || e r || J' \rangle|$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle U e r J' \rangle $ ($e a_0$)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁴ F _{9/2} - 4038 _{9/2}	247.5871	247.5123	1210.8566	1.236	2.040e+07	-	-	-
a ⁴ F _{9/2} - 3994 _{11/2}	250.3739	250.2984	1197.3792	1.879	3.800e+07	-	-	-
a ⁴ F _{9/2} - z ⁴ F _{9/2}	264.0495	263.9708	1135.3647	2.064	4.690e+07	-	-	-
b ⁴ F _{9/2} - 4038 _{9/2}	266.2772	266.1980	1125.8661	1.526	2.500e+07	-	-	-
a ⁴ F _{9/2} - z ⁴ D _{7/2}	266.5578	266.4786	1124.6807	1.518	3.080e+07	-	-	-
b ⁴ F _{9/2} - 3994 _{11/2}	269.5034	269.4234	1112.3886	2.593	5.800e+07	-	-	-
b ⁴ F _{9/2} - 3856 _{7/2}	279.8528	279.7703	1071.2505	1.213	1.700e+07	-	-	-
b ⁴ F _{9/2} - 3822 _{9/2}	282.5276	282.4445	1061.1085	1.668	2.500e+07	-	-	-
a ⁴ F _{3/2} - 3932 _{3/2}	283.7231	283.6397	1056.6374	1.103	1.800e+07	-	-	-
a ⁴ F _{9/2} - z ⁶ G _{9/2}	285.0562	284.9725	1051.6959	1.607	2.260e+07	-	-	-
a ⁴ F _{9/2} - z ⁶ G _{11/2}	292.5646	292.4790	1024.7050	1.588	1.700e+07	-	-	-
a ⁴ F _{7/2} - 4038 _{9/2}	293.5485	293.4627	1021.2706	1.752	2.460e+07	-	-	-
a ⁴ F _{3/2} - z ⁶ F _{1/2}	293.7548	293.6689	1020.5535	0.922	3.400e+07	-	-	-
a ⁴ F _{7/2} - 4029 _{7/2}	294.4010	294.3149	1018.3134	2.105	4.400e+07	-	-	-
a ⁴ F _{7/2} - 3822 _{9/2}	313.4222	313.3314	956.5130	2.644	4.600e+07	-	-	-
a ⁴ F _{9/2} - z ⁶ F _{7/2}	322.1707	322.0776	930.5393	1.613	1.970e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).